HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 09 Name: _____ Marks: / 58

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The polar coordinates of the points X, Y and Z are $(1, 20^\circ)$, $(2, 80^\circ)$ and (r, θ) respectively, where $20^\circ < \theta < 80^\circ$. If $\triangle XYZ$ is an equilateral triangle, find r.

   A. $\sqrt{3}$

   B. $\sqrt{5}$

   C. $\sqrt{7}$

   D. $\sqrt{10}$

(DSE-2025 PII Q24)

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2. The coordinates of the centre of the circle C are $(7,-5)$. If C cuts the x-axis at the points P and Q such that $PQ = 24$, then the equation of C is

A. $x^2+y^2-14x+10y-95=0$.

B. $x^2+y^2+14x-10y-95=0$.

C. $x^2+y^2-14x+10y-119=0$.

D. $x^2+y^2+14x-10y-119=0$.

(DSE-2025 PII Q27)

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3. $3E000000000000_{16} =$

A. $2^{16}+2^{15}+2^{14}+2^{13}+2^{12}$.

B. $2^{17}+2^{16}+2^{15}+2^{14}+2^{13}$.

C. $2^{52}+2^{51}+2^{50}+2^{49}+2^{48}$.

D. $2^{53}+2^{52}+2^{51}+2^{50}+2^{49}$.

(DSE-2025 PII Q31)

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4. It is given that $\log_5 y$ is a linear function of $\log_{25} x$. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 12 and 2 respectively. If $y = mx^n$, then $n =$

A. -12.

B. -4.

C. -3.

D. -1.

(DSE-2025 PII Q33)

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5. Consider the following system of inequalities:

$$\begin{cases} x \leq 11 \\ 4x + 5y - 19 \geq 0 \\ 7x - 6y + 11 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in D, then the greatest value of $8x - 6y + 11$ is

A. 1.

B. 11.

C. 15.

D. 129.

(DSE-2025 PII Q36)

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6. Simplify $\frac{(x^4y^{-5})^3}{xy^2}$ and express your answer with positive indices. (3 marks) (DSE-2025 PI Q01)

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7. The coordinates of the point A are $(4, -2)$. A is rotated clockwise about the origin O through 90° to B.

(a) Write down the coordinates of B.

(b) Suppose that B is translated upwards by t units to the point C. If A, O and C are collinear, find t.

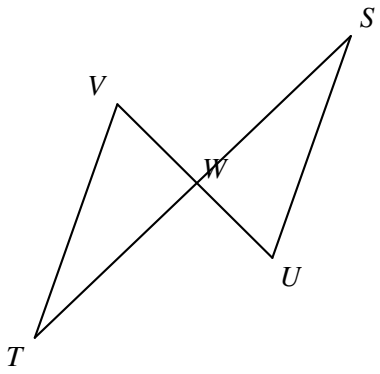
(3 marks) (DSE-2025 PI Q04)

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8. In the figure, ST and UV intersect at the point W. It is given that $SU \parallel VT$ and W is the mid-point of ST.



(a) Prove that $\triangle SUW \cong \triangle TVW$.

(b) Let X be a point lying on TW such that $\triangle SUW \sim \triangle VXW$. If $SU = 57\,\text{cm}$, $SW = 63\,\text{cm}$ and $WX = 7\,\text{cm}$, find the perimeter of $\triangle TVX$.

(5 marks) (DSE-2025 PI Q08)

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9. The table below shows the distribution of the numbers of pens owned by some students.

Number of pens	5	6	7	8	9
Number of students	9	13	2	s	11

The median of the distribution is 7.

(a) Find the least possible value of s.

(b) Write down the greatest possible value of s.

(c) Find the greatest possible standard deviation of the distribution.

(5 marks) (DSE-2025 PI Q09)

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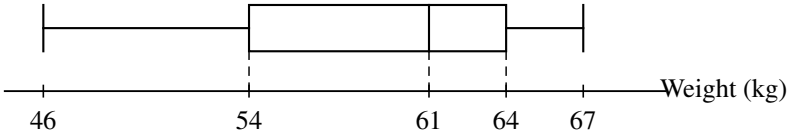
10. The stem-and-leaf diagram below shows the distribution of the weights (in kg) of some athletes before training.

Stem (tens)	Leaf (units)
4	5 6 8
5	w w 6 7
6	3 4 5 6 9 9
7	0 2
8	7

The difference of the range and the inter-quartile range of the above distribution is 25 kg.

(a) Write down the range of the above distribution. Hence, find w.

(b) The box-and-whisker diagram below shows the distribution of the weights (in kg) of the athletes after the training.



(i) Find the change in the upper quartile of the distribution due to the training.

(ii) Is the distribution of the weights of the athletes after the training less dispersed than that before the training? Explain your answer.

(7 marks) (DSE-2025 PI Q12)

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11. There are 8 white plates, 4 red plates and 1 black plate in a box. If 3 plates are randomly drawn from the box at the same time, find

(a) the probability that the colours of the 3 plates drawn are the same;

(b) the probability that the colours of the 3 plates drawn are all different.

(4 marks) (DSE-2025 PI Q15)

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12. Let $g(x) = 3x^2 - 6kx + 24x + 3k^2 - 24k + 55$, where k is a positive constant. Denote the vertex of the graph of $y = g(x)$ by R.

(a) Using the method of completing the square, express the coordinates of R in terms of k.

(b) The graph of $y = h(x)$ is obtained by translating the graph of $y = g(x)$ rightwards by 6 units and then downwards by 10 units. Denote the vertex of the graph of $y = h(x)$ by S. Let T be a point such that the orthocentre of $\triangle RST$ is the origin.

(i) Express the coordinates of T in terms of k.

(ii) The coordinates of the point U are $(-5, 5)$. Let V be a point lying on RS such that TV is perpendicular to RS. It is given that S, T, U and V are concyclic. Find k.

(9 marks) (DSE-2025 PI Q18)

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13. The equation of the straight line L is $4x-3y+83=0$. The equation of the circle C is $x^2+y^2+ax-2y+b=0$, where a and b are positive constants. It is given that C passes through the point $(-10,9)$ and L is a tangent to C.

(a) Prove that the equation $25x^2+(9a+640)x+(90a+4924)=0$ has equal roots. Hence, find a and b.

(b) The coordinates of the point P are $(-20,1)$. Let Q be a point lying below L such that PQ is tangent to C and $PQ=25$. Denote the centre of C by I.

(i) Find $\cos\angle IPQ$.

(ii) Find, in surd form, the distance between I and Q.

(iii) Let R be a point lying on L such that C is the inscribed circle of $\triangle PQR$. Denote the radius of the circumcircle of $\triangle PQR$ by r. Someone claims that $r>PQ$. Is the claim correct? Explain your answer.

(12 marks) (DSE-2025 PI Q19)

I can finish _____ questions

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